

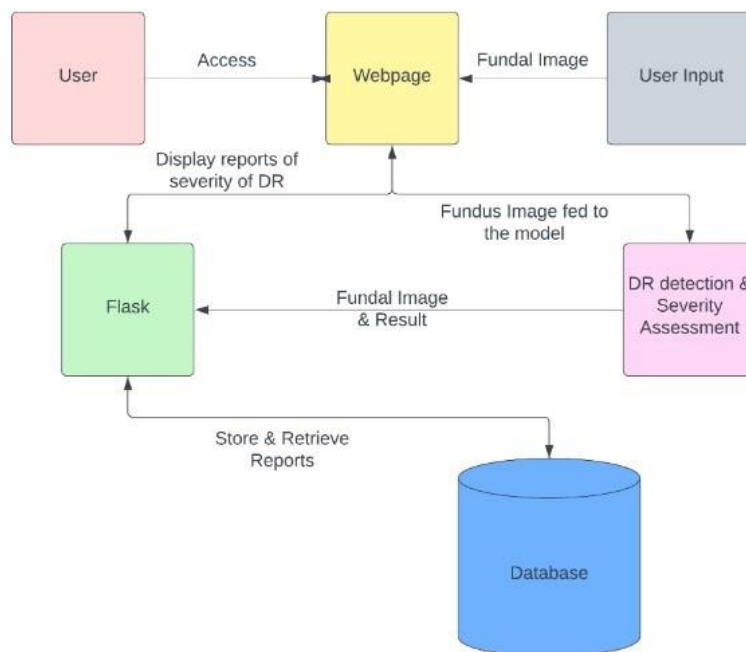
Project Planning Phase - I

Technology Stack (Architecture & Stack)

Date	15 April 2025
Name	Atharv Arun Patil
Project Name	Deep Learning Fundus Image Analysis For Early Detection of Diabetic Retinopathy
Maximum Marks	4 Marks

Technical Architecture:

DIABETIC RETINOPARTHY SEVERITY & REPORTING



1. Users access a webpage to input their fundal image for diabetic retinopathy (DR) detection and severity assessment.
2. The input is processed, and the result along with the fundal image are sent to a Flask application.
3. The Flask app displays reports indicating the severity of DR to the users.
4. The system stores these reports in a database for future retrieval.
5. Users can later access and retrieve their DR reports from the database through the webpage

TABLE-1:

S.No	Component	Description	Technology
1	User Interface	Users interact with the application by submitting retinal images, viewing assessments, and accessing historical data.	HTML, CSS, JavaScript / /React Js
2	Application Logic-1	Handles data preprocessing, executes machine learning models for diabetic retinopathy detection, and generates severity assessments in the web interface.	Python
3	Application Logic-2	Manages user input, custom scenarios in the diabetic retinopathy web interface.	Flask
4	Application Logic-3	Facilitates model performance monitoring and reporting for diabetic retinopathy assessments in the web interface.	Python
5	Database	Stores user data, retinal images, and diabetic retinopathy assessment reports.	MongoDB / PostgreSQL
6	Cloud Database	Provides scalable and secure cloud storage for user data and diabetic retinopathy assessment reports.	AWS
7	File Storage	Storing the model's required files, user data.	Local Filesystem

8	External API-1	Google's OAuth 2.0 authentication service to enable user registration	OAuth 2.0
9.	Machine Learning Model	Utilizes historical data to predict diabetic retinopathy severity, enhancing decision-making within the web interface	Diabetic retinopathy Model(Python).
10.	Infrastructure (Server / Cloud)	Application Deployment on Local System Local Server Configuration:	AWS EC2, Kubernetes, etc.

TABLE-2 :

S.No	Component	Description	Technology
1	Open-Source Frameworks	The open-source Flask framework supports the website, enabling customization and extensibility for precise predictions.	Flask
2	Security Implementations	Utilizes authentication, encryption, and access controls to safeguard website	bitlocker
3	Availability	Can be available through a simple domain (because this project is a web page)	Google Domains,Register.com
4	Performance	Caching: When a user visits a website, the elements get stored in temporary hard drive storage called cache. This process loads pages faster. Reduce HTTP requests: Browsers use a cache to reduce the number and size of HTTP requests, making web pages load faster.	Hard drive (Temporary storage for caching data.)

